MAGNETISM AND MAGNETIC MEASUREMENTS

INTRODUCTION

Magnetism and magnetic materials have a long and illustrious history. Lodestone (a variety of magnetite) was the first magnetic material of technological importance—which often means military importance as well—due to its seemingly mystical power of seeking direction, a feat which never fails to fascinate a first-time observer. Lodestones were known to the ancient Egyptian, Greek, and Chinese civilizations, and possibly others. About 2600 B.C., lodestone was instrumental in winning a decisive battle for the first emperor of China, fought in a dense fog against the invading barbarians.

The subject that we generally call magnetism began with the observation that some minerals, such as magnetite, could attract each other and certain other minerals. The name magnetism probably derives from Magnesia, a location in ancient Thessaly in Asia Minor, presumably where such minerals were found. Despite the early, albeit accidental, discovery of magnetic materials, interest in magnetism during the subsequent centuries was largely limited to the construction of better compasses, more in the realm of witchcraft than science. William Gilbert made the first serious scientific study of magnetism during the 16th century with the publication of *De Magnete*, only to be ignored for more than a century. It was not until the 18th century that scientific interest in electricity and magnetism was renewed with great intensity, culminating with the realization of the Maxwell equations in the 19th century. However, a fundamental understanding of magnetism had to await the advent of quantum mechanics and special relativity in the 20th century.

Today, magnetism enters many aspects of our lives, far beyond just compasses and refrigerator magnets. Magnetic materials and magnetic phenomena inconspicuously appear in many complex devices, for example, theft-prevention systems in department stores, magnetic recording disks and read heads in the ubiquitous computers and cellular phones, and in magnetic resonance imaging apparatus. Some of the more exciting new developments are the spintronic devices, which are on the verge of becoming realities. These new devices are based on the spin properties of electrons, as opposed to charge alone, as in conventional electronics.

The subject of magnetism is inseparable from electrical charges, their translational motion and their angular momenta. Magnetic fields can be generated by current-carrying wires, through which the electrical charges (most commonly the electrons) are transported. Magnetic fields can also be realized in the vicinity of magnetic materials with large magnetization, such as permanent magnets. There are various methods by which one generates magnetic fields, ranging from generation of magnetic fields by electrical circuits and permanent magnets to the use of hybrid superconducting magnets, Bitter magnets, and pulsed magnets, necessary for achieving very large magnetic fields. Roughly speaking, magnetic fields up to about 2 tesla (20 kOe) and 10 tesla (100 kOe) can be generated by laboratory electromagnets and superconducting magnets, respectively. There are also national facilities, such as the National High Magnetic Field Laboratory at Florida State University, which accommodate measurements at much higher magnetic fields.

All constituent atoms in a material respond to an external magnetic field, at least weakly due to diamagnetism. The direction of the small diamagnetically induced magnetization is opposite to that of the external magnetic field. However, virtually all magnetic materials of scientific or technological importance are those with sizable atomic moments, which respond strongly to external magnetic fields. The atomic magnetic moments are the results of orbital and spin angular momenta of the electrons in the partially filled atomic electronic shells. These atomic magnetic moments, with magnitude of the order of Bohr magneton (μ_B), are much larger than those induced by diamagnetism. Most of the elements in the periodic table do not have atomic magnetic moments, except some of the 3*d* transition-metals (e.g., Ni, Fe, Co, and Mn) and 4*f* rare-earth metals (e.g., Gd, Nd, etc.), which are the major constituents of all the important magnetic materials.

The atomic magnetic moments can interact with each other via the quantum mechanical exchange interactions. At high temperatures, the atomic moments are not ordered and are in the paramagnetic state. However, at sufficiently low temperatures, the atomic moments can spontaneously order. Those materials in which the atomic moments align parallel to each other are the ferromagnets, of which lodestones, refrigerator magnets, and other more powerful permanent magnets are examples. Those materials that exhibit nonaligned spin structures (e.g., antiferromagnetic and various other types of spin structures) are technologically not useful but scientifically fascinating. Each type of magnetic ordering has a unique order parameter, which, for example, is the magnetization for ferromagnets and sublattice magnetization for antiferromagnets. Various methods of magnetometry can be used to measure the magnetization of materials and the atomic magnetic moments. Neutron diffraction is particularly valuable for measuring the spin structure of magnetic ordering, particularly antiferromagnets and other more complex magnetic orderings, as described in NEUTRON TECHNIQUES.

Most people recognize that a piece of iron can easily be attracted by a permanent magnet, but iron itself cannot attract another piece of iron. This is because in iron there are magnetic domains, a general phenomenon in ferromagnets. Under a sufficiently large magnetic field, a

ferromagnetic material can always be swept into a single-domain state, where all the moments are aligned in one direction. In a small or zero magnetic field, many magnetic domains, separated by domain walls, may form in order to reduce the magnetostatic energy. Within each domain, all the moments remain aligned with a unique magnetization, but the magnetization axes among different domains are not aligned. The size and shape of the domains depend on many factors, most importantly the magnetic anisotropy. In a soft magnet, such as pure iron, domains can be readily formed and swept by a small field because of the small magnetic anisotropy. In a hard magnet, such as the refrigerator magnets and other more powerful permanent magnets, the magnetic anisotropy is so strong that it preserves the magnetization within the material. The magnetic domains, as small as a few tens of nanometers in size, have fascinating shapes and patterns. A variety of techniques have been developed to image magnetic domains and study the domain wall motion.

Electrons have both spin and charge. Many of the magnetic materials are electrical conductors. When electrons traverse magnetic materials, the external field and the magnetization **M** of the magnetic medium affect the conductivity. These are magneto-transport properties, which include a variety of interesting topics, such as magnetoresistance and the Hall effect, and the latter is described in ELECTRICAL AND ELECTRONIC MEASUREMENT. New magneto-transport properties due to spin-dependent scattering as well as spin-polarized conduction have recently been discovered in suitable magnetic heterostructures. These new effects form the basis for some of the new spintronic devices.

Most of the magnetic materials of technological importance are ferromagnets. These include permanent magnets (e.g., ferrite, $SmCo_5$, $Nd_2Fe_{14}B$) that can generate strong magnetic fields, soft magnetic materials (e.g., permalloy, amorphous ferromagnets) that can respond to small magnetic fields, materials that have large magneto-transport properties, and materials that can store magnetization, as in magnetic recording media (e.g., γ-Fe_2O_3, CoTa alloys).

The smallest magnetic entities are magnetic dipoles. Under a uniform magnetic field, a magnetic dipole of dipole moment $\boldsymbol{\mu}$ can only rotate. Under a nonuniform magnetic field **H**, the magnetic dipole can also have translational motion due to the force arising from the field gradient, $\mathbf{F}_\mu = (\mu \cdot \nabla)\mathbf{H}$. For a material of magnetization **M**, the total force per unit volume is $\mathbf{F}_v = (\mathbf{M} \cdot \nabla)\mathbf{H}$. The force on a ferromagnet is attractive and substantial, as evidenced by the refrigerator magnet. For nonmagnetic materials, the force is repulsive due to diamagnetism, and therefore usually weak. However, in an intense magnetic field gradient, the repulsive force can be quite substantial, so much so that it can actually lift or levitate an object against gravity.

Thus far, we have mentioned the magnetic properties due to the electrons in the atoms and the materials. We have ignored the nuclei and other particles because their magnetic moments are three orders of magnitude smaller. However, in certain phenomena, such as nuclear magnetic resonance, Mössbauer effect, muon spin resonance, and neutron scattering, these small magnetic moments are the only ones of importance as described in RESONANCE METHODS.

Units

The descriptions of magnetic properties must include the discussion of units. We are all supposed to be, and indeed have been, using the SI units (Systéme Internationale d'Unités). However, one notable exception is in magnetism, where the systems of units have been confusing and challenging even for seasoned practitioners. Although all the more recent physics textbooks have used exclusively the SI units to describe magnetic quantities as they should, most scientific literature in magnetism, both old and new, continues to use the cgs units, or a mixture of both cgs and SI units. The cgs units also persist on the instrument side. Most commercial magnetometers display magnetic moment in emu (electromagnetic unit) and most gaussmeters measure magnetic field in Oe (Oersted). A practitioner in magnetism therefore must know both the SI and the cgs units.

In the cgs units, the relation among magnetic induction or magnetic flux density (**B**), magnetic field intensity (**H**), and magnetization (**M**) is uniquely defined as

$$\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M} \tag{1}$$

This expression is universally used without ambiguity. Since the units for B (gauss = G), H (oersted = Oe), and M (emu/cm^3) have the same dimension, sloppiness in describing the physical quantities is therefore tolerated (lightface symbols represent the scalar magnitude of the corresponding vector quantity). For example, one can describe the earth's magnetic field as about 0.5 G or 0.5 Oe, and the magnetization of a piece of iron can be specified as M = 1700 emu/cm^3 or 4πM = 21.7 kG.

In the SI units, the corresponding relation is

$$\mathbf{B} = \mu_o(\mathbf{H} + \mathbf{M}) \tag{2}$$

where μ_o is a fundamental constant of $4\pi \times 10^{-7}$ N/A^2. The units for H and M are in A/m, whereas B is in tesla, or Wb/m^2. Most unfortunately, other forms, such as $\mathbf{B} = \mu_o\mathbf{H} + \mathbf{M}$, have also been used, creating endless confusion.

In Table 1, various common quantities used in magnetism are shown both in the cgs units and the SI units, as well as the conversion factors.

MAGNETIC SUSCEPTIBILITY AND PERMEABILITY

As described earlier, the applied external magnetic field, **H**, the magnetization, **M**, and the magnetic flux density, **B**, inside a medium are related by Equation 1 in cgs units and Equation 2 in SI units. An external field, **H**, induces certain magnetization, **M**, in a sample, and as a result, generates an internal flux density, **B**. We will use the cgs units in the discussion below for brevity. One can easily convert values into SI units by referring to Table 1.

The magnetic response of a medium to an external field is often characterized by two related parameters of

Table 1. Conversion Between cgs Units and SI Units for Selected Magnetic Quantities

Quantity	Symbol	cgs Units ($\mathbf{B} = \mathrm{H} + 4\pi\mathbf{M}$)	Conversion Factor (C) [(SI) = C(cgs)]	SI Units [$\mathbf{B} = \mu 0(\mathbf{H} + \mathbf{M})$] [$\mu 0/4\pi = 10-7\mathrm{N/A2}$]
Magnetic induction	B	Gauss (G)	10^{-4}	tesla(T) Wb/m^2
Magnetic flux	Φ	G-cm^2 Maxwell(Mx)	10^{-8}	weber(Wb)
Magnetic field intensity	H	Oersted(Oe)	$10^3/4\pi$	A/m
Magnetic moment	m	emu, erg/Oe	10^{-3}	A-m^2, J/T
Magnetization	M	emu/cm^3	10^3	A/m
	4πM	G	$10^3/4\pi$	A/m
Mass magnetization	M_g	emu/g	1	A-m^2/kg
			$4\pi \times 10^{-7}$	Wb-m/kg
Volume susceptibility	χ	Dimensionless	4π	dimensionless
Mass susceptibility	χ_g	cm^3/g	$4\pi \times 10^{-3}$	m^3/kg
Permeability	μ	Dimensionless	$4\pi \times 10^{-7}$	Wb/A-m

magnetic susceptibility (χ) and permeability (μ), which are defined as

$$\chi = \mathrm{M}/\mathrm{H} \tag{3}$$

$$\mu = \mathrm{B}/\mathrm{H} \tag{4}$$

and $\mu = 1 + 4\pi\chi$. If the magnetization of a medium is proportional to H, χ is then a constant and becomes the most appropriate parameter to characterize the magnetic response. On the other hand, if M is not linear in H, χ and μ become functions of H, i.e., $\chi = \chi(\mathrm{H})$, $\mu = \mu(\mathrm{H})$. In some situations, such as the case of a ferromagnet, the behavior of M is hysteretic and, therefore, not uniquely determined by a given H. The concepts of χ and μ are no longer very useful, although still adopted, particularly in soft magnetic materials.

PARAMAGNETISM

A paramagnet consists of an assembly of magnetic dipole moments—μ—the atomic magnetic moments. Thermal disturbance randomizes the orientations of these moments. An external field, on the other hand, creates a tendency for a slight alignment of the moments along the field direction. Under the condition of $\mu\mathrm{H} \ll kT$, the induced **M** is proportional to **H**. If the interaction between the moments is negligible, the susceptibility is given by the Curie Law

$$\chi = C/T = N\mu_{\mathrm{B}}^2 p_{\mathrm{eff}}^2/3kT \tag{5}$$

where C is the Curie constant, N is the number of magnetic moments per unit volume, $\mu_{\mathrm{B}} = e\hbar/2mc$ is the Bohr magneton, and p_{eff} is the effective moment number in μ_{B}. Curie's Law offers an effective means to determine the magnitude of atomic or molecular magnetic moment in a system. Or, if the moment of interest is known, one can also determine the concentration of the magnetic component. In experiments, one often plots $1/\chi$ versus T. The slope of the straight line provides the information needed.

In many cases, there may exist some sort of interaction between the individual moments. The temperature dependence of χ is then described by the Curie-Weiss Law

$$\chi = C/(T - \theta) \tag{6}$$

where the constant θ is positive (negative) for interaction of ferromagnetic (antiferromagnetic) origin. Equation 6 can be used to determine θ, which is close to the ferromagnetic ordering temperature T_{c}.

The Curie or the Curie-Weiss Law becomes invalid when the condition of $\mu\mathrm{H} \ll kT$ is not met. This could happen when H is appreciable, or the magnetic moment (μ) of interest is large. The latter is particular relevant for a superparamagnetic system, which consists of a collection of ferromagnetic or ferrimagnetic particles with moments much larger than constituent atomic moments. Under these situations, M becomes the Brillouin function of H, and χ is field dependent.

A nonmagnetic metal or alloy can exhibit a positive and essentially temperature-independent susceptibility when subjected to a field. This phenomenon, called Pauli paramagnetism, is caused by the Zeeman splitting of the spin-up and spin-down conduction bands induced by the field. A thermal equilibrium process leads to the redistribution of the two types of electrons, leaving the system with a net magnetization. Pauli paramagnetic susceptibility is given by

$$\chi = \mu_{\mathrm{B}}^2 D(E_{\mathrm{F}}) = 3N\mu_{\mathrm{B}}^2/2E_{\mathrm{F}} \tag{7}$$

where $D(E_{\mathrm{F}})$ is the density of states at the Fermi level (E_{F}). It should be pointed out that Pauli susceptibility is at least one order of magnitude smaller than the paramagnetic susceptibility of magnetic ions. Discrepancy is often found between experiments and the predicted values according to Equation 7, which is derived from the free electron gas model. In reality, electron-electron or other types of interaction may not be omitted.

Another weak and temperature-independent susceptibility can be obtained from the second-order Zeeman effect, as was first proposed by van Vleck. This susceptibility depends on the energies of the low-lying exited states

$$\chi = 2N \sum |\langle \alpha_i | \mu_z | 0 \rangle|^2 / (E_{\alpha i} - E_0) \qquad (8)$$

where $\langle \alpha_i |$ and $|0\rangle$ represent the excited and the ground states.

The total susceptibility of a material is the sum of Equation 6, Equation 7, and Equation 8, plus the contribution from a negative susceptibility due to diamagnetism.

DIAMAGNETISM

According to Lenz's law of electromagnetism, application of an external field on an electron system induces an electrical current that will generate a field opposing the change of the external field. The microscopic current within an atom or molecule is dissipationless, and will create a magnetic moment vector opposite and proportional to the external field. This so-called diamagnetic susceptibility is a negative constant, given by

$$\chi = -(Ne^2/6mc^2) \sum \langle 0 | r_i^2 | 0 \rangle \qquad (9)$$

where r_i is the electron radius. Diamagnetism exists in all materials. Its susceptibility is small in comparison with Curie paramagnetism. It does not depend on temperature, and is proportional to the total number of electrons in an ion or a molecule. The temperature dependence of various paramagnetic and diamagnetic susceptibilities is shown schematically in Figure 1.

A superconductor displays a unique diamagnetism. The induced supercurrent on the surface of a type I superconductor suppresses completely the magnetic flux density (i.e., $\mathbf{B} = 0$) within the body of the superconductor. This perfect diamagnetism is called the Meissner effect. The susceptibility is, according to Equation 1, $\chi = -(1/4\pi)$ emu/cm^3, which is many orders of magnitude larger than that of ordinary diamagnets. In a type II superconductor, a magnetic field can partially penetrate into the superconductor. Therefore, the susceptibility is smaller than $1/4\pi$ emu/cm^3 in magnitude, and is dependent on the field strength.

Because of the negative susceptibility, there is a repulsive force between any diamagnetic material and the source of the external nonuniform field. A levitating superconductor on a magnet is a famous example for such repulsion. Researchers have also used a magnetic field to levitate liquid helium droplets (Weilert et al., 1997) and biological materials (Valles et al., 1997).

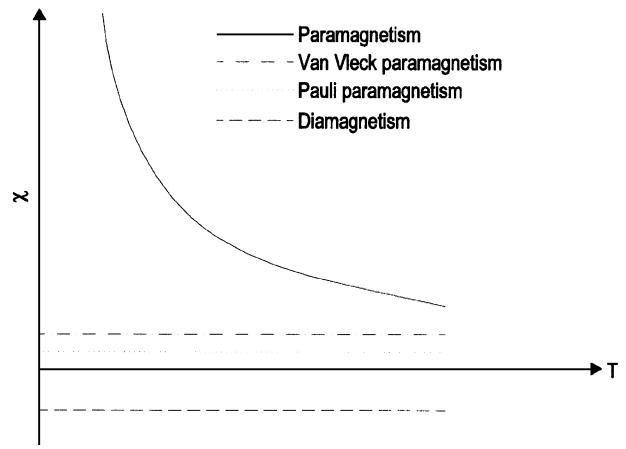


Figure 1. Temperature dependence of susceptibility for paramagnetic and diamagnetic systems.

ANTIFERROMAGNETISM

In an antiferromagnet, magnetic moments at adjacent atomic sites or planes develop a spontaneous antiparallel alignment below a critical phase transition temperature, T_N. The physical mechanism of antiferromagnetism is due to the exchange interaction, which originates from the many-body Coulomb interactions among electrons. Above T_N, the long-range order is destroyed, and the susceptibility is given by the Curie-Weiss Law (Equation 6) with a negative θ. Below T_N, the net magnetization is zero in the absence of an external field, because of the antiparallel configurations of the moments. An external field will induce spin canting and generate a net magnetization, which is proportional to H. In general, there are two susceptibilities, $\chi_\perp$ and $\chi_{||}$, corresponding to the external field perpendicular and parallel to the axis of the spins (easy-axis) respectively.

Figure 2 shows schematically the temperature dependence of susceptibility of an antiferromagnetic crystal. Below T_N, $\chi(T)$ changes abruptly from the Curie-Weiss behavior and branches into $\chi_\perp(T)$ and $\chi_{||}(T)$. As T approaches 0 K, $\chi_{||}$ tends to vanish rapidly. The quantity $\chi_\perp$, on the other hand, remains constant or weakly dependent on T. For polycrystalline or powder materials, $\chi(T)$ falls between $\chi_\perp(T)$ and $\chi_{||}(T)$ as a result of averaging.

FERROMAGNETISM

Ferromagnetic solids are an important class of materials used in a wide range of applications. Many magnetometry

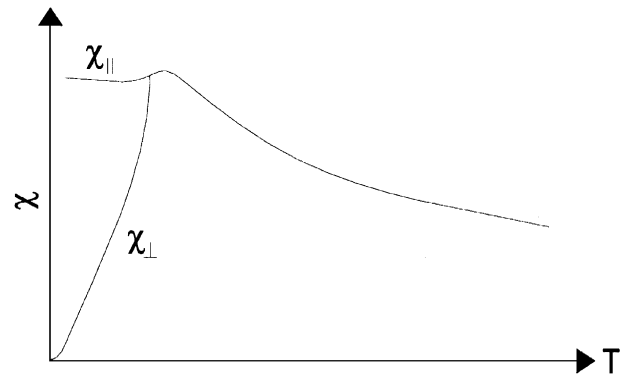


Figure 2. Temperature dependence of susceptibility for an antiferromagnetic system.

techniques have been developed to study the properties of these solids. With exchange interaction as its physical origin, ferromagnetism exhibits a long-range ordering of parallel magnetic moments below a second-order phase transition temperature T_c. Above T_c, the magnetic behavior can be described by the Curie-Weiss Law (Equation 6). From the slope [i.e., the constant C in Equation 5 and Equation 6 of the straight line $1/\chi(T)$ versus T], one can deduce the size of the atomic magnetic moment.

Below T_c, the magnetization is not uniquely dependent on external field H, but rather is determined by the magnetic history of the sample. Experimentally, one either plots M or B versus a cycling H. These plots are called the magnetic hysteresis loops. Since in a ferromagnet M tends to be much larger than H, the two loops (M versus H and B versus H) look similar. Figure 3 shows a schematic hysteresis loop of M versus H and the initial magnetization curve. In the unmagnetized state, the net M is zero because of the magnetic domain structure. As H is increased, M follows the initial magnetization curve until magnetic saturation is reached. Afterwards, the cycling of magnetic field between a positive and negative value causes the M to form a closed loop.

There are a few parameters to characterize the hysteresis loop. The saturation magnetization M_s corresponds to a complete parallel alignment of magnetic moments. The remnant magnetization M_r is the nonvanishing value when H is reduced to zero. The coercivity H_c is the reversing field needed to bring M to zero. The area enclosed by the hysteresis loop is related to the so-called energy product, which is a measure of the prowess of a permanent magnet.

For a permanent magnet, the larger the M_s, M_r, and H_c, the better. In other words, a loop with a large area and a large energy product is preferred. Because of the hysteretic nature of the magnetization process, the definitions of χ and μ according to Equation 3 and Equation 4 are not useful.

For a soft magnet, the H_c is in general very small ($\sim 10^{-2}$ Oe), and so is the loop area. Because of their low energy loss, soft magnetic materials are extensively used in applications involving alternating fields, such as transformers and RF inductors. One often uses modified definitions of susceptibility and permeability to characterize a soft magnetic material. Efforts have been made to enhance permeability for superior performance.

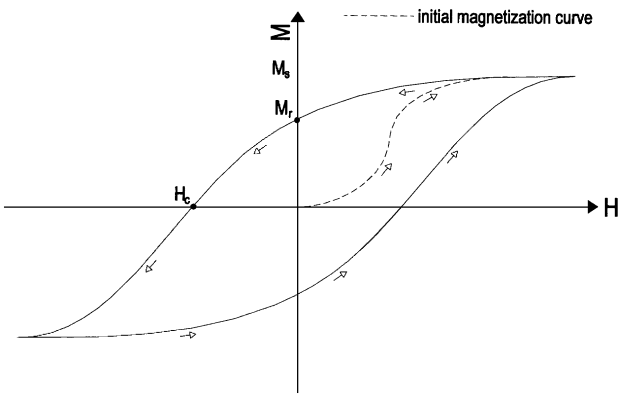


Figure 3. Hysteresis loop for a ferromagnetic system.

In some applications, such as in RF systems, the magnetic field scale is small. The relevant parameter is the initial permeability, defined as

$$\mu = (dB/dH)_{H=0} \tag{10}$$

As **H** increases, $\mu = B/H$ will increase and reach a certain maximum value, which is called maximum permeability μ_M. In some pure and well annealed soft magnetic materials, μ_M can be as large as 10^6. Often permeability data given in the literature refers to μ_M. At each given field, one can also specify a differential permeability $\mu = (dB/dH)_H$.

If a magnet is operated under a bias field ($\mathbf{H}_0$) plus a small alternating field ($\Delta\mathbf{H}$) around $\mathbf{H}_0$, **M**, and **B** will form a minor loop. In this case, one can also define an incremental permeability $\mu_D = (\Delta B/\Delta H)_H$.

When μ is large, as in a typical soft magnetic solid, χ is approximately proportional to $\mu(= 1 + 4\pi\chi \approx 4\pi\chi)$. The aforementioned definitions of various permeabilities can be extended to the corresponding susceptibilities.

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GENERATION AND MEASUREMENT OF MAGNETIC FIELDS

INTRODUCTION

In basic science areas, such as semiconductor physics, superconductivity, structural biology, and chemistry, magnetic fields are used as a powerful tool for analysis of matter because they can change the physical state of a

system. In the presence of a magnetic field, the angular momenta (spins), the orbital motion of charged particles, and therefore the energy of the investigated physical system, are altered. We must realize, however, that the size of the induced effects are very modest compared to other physical quantities. One Bohr magneton (typical range of atomic magnetic moments) in a magnetic field of 10 tesla (T) (a 10 T field is ~200,000 times the strength of the earth's magnetic field) has an energy of 0.58 meV, corresponding to a temperature of 6.7 K, a small value in comparison to room temperature. On the other hand, the generation of a magnetic field of 10 T already requires a noticeable engineering effort, with continuous fields of 50 T being the upper limit of what can be achieved today. Pulsed magnetic fields can be generated up to 100 T for durations in the millisecond range, and up to 1000 T on the microsecond time scale. Therefore, experiments must be chosen judiciously. Very often, a low-noise sample environment, such as low mechanical vibrations, small field fluctuations (in the case of dc fields), and low temperatures are decisive to the success of scientific measurements in magnetic fields.

EFFECTS AND APPLICATIONS OF MAGNETIC FIELDS

The use of magnetic fields and their effects on matter are based on a few micro- and macroscopic principles. These principles are utilized not only in basic science areas but also for significant industrial applications (Schneider-Muntau, 1997), and most of them are also employed to measure magnetic fields. These include the following.

1. Magnetization of nuclei and its decay. This effect is the basis for nuclear magnetic resonance (NMR) and magnetic resonance imaging (MRI; NUCLEAR MAGNETIC RESONANCE IMAGING). A considerable industry has developed with numerous applications: NMR spectroscopy, imaging ("scanner") with relevance in medical science, structural biology, and pharmaceutical development, but also for the determination of material properties and nondestructive process and quality control. NMR allows the most accurate measurement of magnetic fields.
2. Induced currents. The electromagnetic induction, i.e., the interaction between conductors and magnetic fields, gives rise to many effects, the most significant applications of which are the generator and electromotor. Magnetic levitation of trains, magnetic propulsion, and magnetic damping of conductive liquids for crystal growth are based on this same effect. Rotating coil gauss meters and extraction flux integrators represent easy ways to measure magnetic fields. Pick-up coils are a standard method for the measurement of pulsed fields.
3. Force on moving charged particles ("Lorentz force"). Accelerators and detectors for particle physics, plasma confinement in fusion devices, isotope separation, ion cyclotron resonance mass spectrometers, and electron beam deflections in TV sets and computer monitors are the most significant applications of this effect. The Lorentz force is, as well, the primary limiting factor in the generation of high magnetic fields, since this force can easily exceed the mechanical strength of the conductors of an electromagnet. The Hall effect (see DEEP-LEVEL TRANSIENT SPECTROSCOPY) is widely implemented for sensors in industry, and it is also the most common device for the measurement of magnetic fields.
4. Magnetization. In a magnetic field, matter is polarized and its magnetic anisotropy results in a force. This effect is employed for separation, alignment of crystals, and levitation. Today's high-field magnets are so powerful that most diamagnetic matter, for instance water and therefore small animals and plants, can be suspended.
5. Energy content. The energy stored in a magnetic field can be effectively utilized in industrial applications. Small systems are available for improvement of the quality of electrical power. Several demonstration projects in the 10-kJ/10-kW range have proven the feasibility of magnets as energy storage systems.

GENERATION OF MAGNETIC FIELDS

Magnetic fields can be generated by magnets created from hard magnetic materials and by electromagnets. Permanent magnets offer a simple, fast, and cost-effective way to provide small-volume fields. *Permanent magnets* are being employed for MRI applications, accelerator storage rings, and wiggler magnets. Fields up to 4 T have been achieved. Devices exist that can produce variable fields or fields with changing orientation. The basic principles of permanent magnets and some applications are described below.

Permanent magnets can also be made from bulk high-temperature superconductors, such as $YBa_2Cu_3O_7$ single crystals. In recent developments it has been shown that a magnetic field of 10 T can be maintained in a mini-magnet of 2-cm diameter, cooled to 42 K (Weinstein et al., 1997).

High and very high fields can only be generated with *electromagnets*. First, some basic principles common to all electromagnets are described, followed by a more detailed treatment of *superconducting magnets*, *resistive magnets*, and *pulse magnets*.

Generation of the highest fields possible is limited by the strength of materials and, for resistive magnets, by the available energy or power source. Because of the resulting magnet size and therefore cost, as well as the required human resources and technical infrastructure, on one hand, and the need for high magnetic fields for research and technology developments, on the other hand, governments have established national facilities as central institutions to serve the scientific community. We provide a short overview of the most important *magnet laboratories* around the world. Each of these laboratories entertains an intensive user program, and access is usually free of charge.

Permanent Magnets

Modern permanent magnets are ideally suited to generate magnetic fields of magnitudes comparable to their spontaneous polarization J_s. Polarization (J), in tesla, is related to magnetization (M) in A/m by the equation $J = \mu_0 M$, where $\mu_0 = 4\pi \cdot 10^{-7}$ Tm/A. In the centimeter-gram-second (cgs) system, M is measured in emu/cm^3 (1 emu/cm^3 = 1 kA/m), $\mu_0 = 1$, and the polarization $4\pi M$ is measured in gauss (1 G = 10^{-4} T); the cgs energy product is measured in megagauss-oersteds (1 MGOe $\approx$ 8 kJ/m^3). The fields can be static or variable, uniform or nonuniform. They are generated by compact permanent magnet structures requiring no continuing expenditure of energy for their operation. The main classes of hard magnets are based on hexagonal ferrites, such as $BaFe_{12}O_{19}$, for which the intrinsic polarization of the pure phase is $J_s = 0.5$ T, and rare earth intermetallic compounds such as $Nd_2Fe_{14}B$, for which $J_s = 1.6$ T. Permanent magnets are now fully competitive with electromagnets for generating fields of up to $\sim$2 T; indeed when a field with a rapid spatial variation is required, they may offer the only practical solution.

Ferromagnetic materials normally have a multidomain structure below their Curie point. The atomic moments within each domain are aligned along some easy direction; the electric currents associated with the quantized spin and orbital moments of the electrons add to produce a net circulating current density of order 1 MA/m, which corresponds to the spontaneous magnetization M_s (1 MA/m $\equiv$ 1.26 T). In the unpolarized state, the moments of the different domains are aligned in different directions and the net magnetization is zero. The domain structure is eliminated by applying an external field, H, which is sufficiently large to align the magnetization of each domain and saturate polarization (Fig. 1). What happens when the field is removed depends on the nature of the material. In a soft magnetic material, the domain structure reforms as soon as the applied field is less than the local demagnetizing field, and hysteresis is negligible; in a hard magnetic material, the polarization remains close to its saturation value well into the second quadrant of the hysteresis loop, and falls to zero at a negative value known as the coercive field, H_c. The hysteresis loop is broad, and for a true permanent magnet the coercive field should be $>M_s/2$.

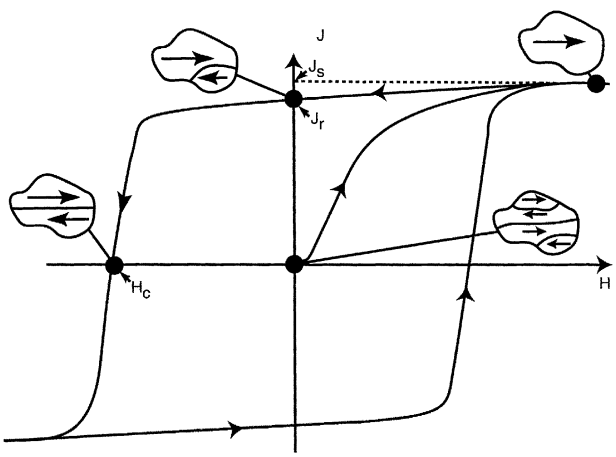


Figure 1. The hysteresis loop of a permanent magnet.

The fully magnetized state is inherently metastable and tends to revert to the equilibrium, multidomain configuration—a process that must be indefinitely delayed in practical magnets. There are two main strategies for achieving this. Permanent magnets are generally constituted of a main phase which has a uniaxial crystal structure (rhombohedral, tetragonal, hexagonal) and strong easy-axis anisotropy so that the magnetization can lie only along the c axis. The anisotropy energy is often represented by an equivalent anisotropy field, B_a. There is, therefore, an anisotropy energy barrier to magnetization reversal. Where the barrier is lowest, a reverse domain may nucleate, and eventually the domain wall may sweep across a particle, completely reversing its magnetization. One approach is to inhibit nucleation of reverse domains by producing a microstructure of tiny ($\sim$1 to 5 μm) crystallites with smooth surfaces, possibly separated by a nonmagnetic secondary phase. The hexagonal ferrites, $SmCo_5$ and $Nd_2Fe_{14}B$ are nucleation-type magnets. The other is to pin any domain walls by local defects associated with nanoscale inhomogeneity in the magnetic material. Sm_2Co_{17} is a pinning-type magnet. The decay of the permanent moment with the logarithm of time is an effect known as magnetic viscosity. It may be of the order of 1% in the first year after magnetizing the material, and cannot be avoided. The medium-term irreversible losses can, however, be exhausted by thermal aging.

The *energy product* $(BH)_{max}$ is the usual figure of merit for a permanent magnet; it corresponds to twice the maximum energy that can be stored in the magnetic field produced around an optimally shaped magnet. In order to maximize the polarization and energy product, the individual crystallites should be aligned with their c axes in a common direction. Details of the intrinsic magnetic properties (Curie temperature, T_C, spontaneous polarization, J_s, and anisotropy field, B_a) of a range of modern permanent magnet materials are listed in Table 1, together with the properties of typical magnets made from them—coercive field, H_c, remanent polarization, J_r, and energy product, $(BH)_{max}$. The polarization of the magnet is normally less than that of the hard phase because of imperfect crystallite alignment, or the presence of nonmagnetic secondary phases needed to achieve coercivity.

All of these magnets can be regarded as transparent, to a first approximation. This means that the polarization of any block of magnetic material is unaffected by the presence of other magnet blocks in its vicinity. Transparency greatly simplifies the design of magnetic circuits. The flux produced by an arrangement of permanent magnets at a point $\mathbf{r}$ is then the product of a geometric factor $f(\mathbf{r})$ and the polarization of the magnets.

The flux density, B, produced by a small magnet of moment $m = MV$ Am^2 is nonuniform and anisotropic. Here $M = J/\mu_0$ is the magnetization, μ_0 is the free space permeability, and V is the magnet volume. The two perpendicular components of the field produced in the plane of a small dipole are

$$B_r = (\mu_o m/4\pi r^3)2\cos\theta \quad \text{and} \quad B_\theta = (\mu_o m/4\pi r^3)\sin\theta \quad (1)$$

Table 1. Properties of Permanent Magnet Materials

Compound	Structure	T_C (K)	J_s (T)	B_a (T)	H_c (MA/m)	J_r (T)	$(BH)_{max}$ (kJ/m^3)
$BaFe_{12}O_{19}$	Hexagonal	723	0.48	1.65	0.28	0.39	28
$SmCo_5$	Hexagonal	993	1.05	40	1.40	0.88	150
Sm_2Co_{17}	Rhombohedral	1100	1.30	6.5	0.90	1.08	220
$Nd_2Fe_{14}B$	Tetragonal	585	1.61	7.6	1.00	1.25	300

In order to generate a uniform field over some region of space, segments magnetized in different directions need to be assembled. An effective way of doing this is to build the structure from long segments with a moment l Am. In this case, the field is given by

$$B_r = (\mu_o l/4\pi r^2)2\cos\theta \quad \text{and} \quad B_\theta = (\mu_o l/4\pi r^2)2\sin\theta \quad (2)$$

so that the magnitude

$$|B| = \sqrt{B_r^2 + B_\theta^2} \quad (3)$$

is now independent of θ, the angle between the radius vector and the magnetization. The dipole ring shown in Figure 2A is an ideal structure where the direction of magnetization in the magnet is given by $\alpha = 2\phi$, with $\phi = \pi - \theta$. The field generated is uniform within the ring and zero outside, provided the ring is very long. The value B_i of the field in the bore for a material with an ideal square hysteresis loop (J_S is the spontaneous magnetization) is

$$B_i = J_s \ln(r_2/r_1) \quad (4)$$

The device illustrates nicely the principle of flux concentration. For example the field generated at the end of a long bar magnet is just J_r, but larger values may be achieved in a magnetic circuit containing segments magnetized perpendicular to the field direction in the airgap. To obtain a field equal to the polarization, we need $r_2/r_1 = 2.7$, but a field of twice J_s needs a radius ratio of 7.4 and 20 times as much magnetic material. Even if materials existed with sufficient coercivity to withstand the large reverse fields acting on some regions of the structure, permanent magnets required to produce multi-tesla fields soon become impractically large.

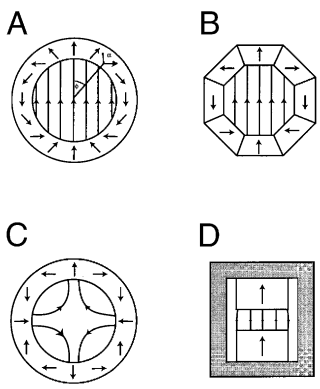


Figure 2. Permanent magnet structures used to generate (**A**) a uniform field, and (**C**) a quadrupole field. The structure shown in (**B**) is a simplified version of (**A**), and (**D**) is a different design, using a soft iron yoke (shaded).

The ring structures can be used to generate higher multiple fields with a different ratio of α and ϕ; for a quadrupole field, for example (Fig. 2C), $\alpha = 3\phi$.

The rings of Figure 2A and C are difficult to realize in practice, and simplified versions with their length comparable to their diameter can be constructed from trapezoidal segments (Fig. 2B). Other designs composed of prismatic segments generate uniform fields in rectangular or triangular bores. It may be advantageous to include soft material in the design to provide a return path for stray flux, as shown in the yoked design of Figure 2D.

These permanent magnet structures are also useful when a rotating magnetic field is needed, or when it is necessary to toggle a field between two directions differing by a fixed angle (such as 180°), in order to reverse the field.

The task of generating a uniform variable field may be undertaken with permanent magnets, provided some movement is permitted. A useful design (Leupold, 1996) consists of two nested dipole rings. The one illustrated in Figure 3A uses cylinders of the type shown in Figure 2B. Dimensions are chosen so that each cylinder produces

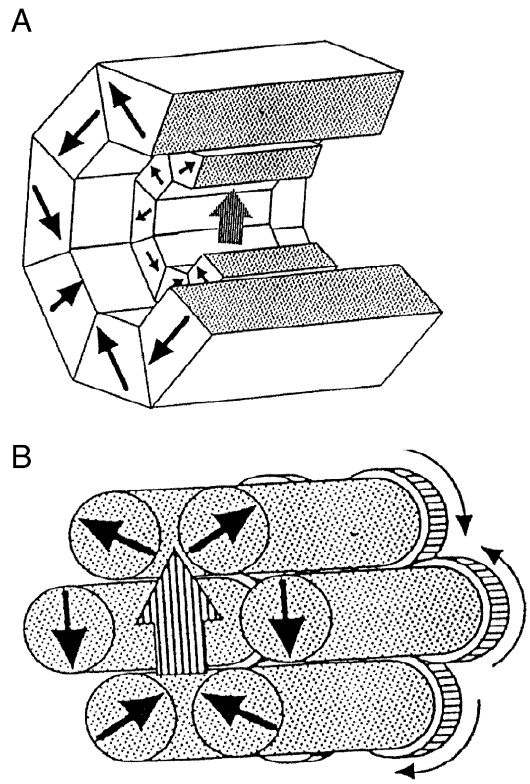


Figure 3. Practical permanent-magnet structures for generating variable fields. (**A**) A double cylinder ("magic cylinder"), and (**B**) a 4-rod design ("magic mangle").

the same field in its bore, and they can be independently rotated about their common axis. In this way, a field in the range $-2B_i < B < 2B_i$ can be produced in any direction in the plane perpendicular to the common axis of the cylinders. The same device can also generate a rotating field of any magnitude from zero up to $2B_i$. When access parallel to the field is desired, a different design, based on rotating cylindrical rods, is preferred. One with six rods is shown in Figure 3B. With four rods, the maximum field available is just J_s (Coey and Cugat, 1994).

There are innumerable applications of permanent magnets as flux sources. They are extensively used in small dc motors and actuators. One example, which accounts for a significant fraction of the global production of $Nd_2Fe_{14}B$, is the voice coil actuator that moves the read head across the surface of a hard disc. Static fields produced by permanent magnets are also used for magnetic separation, magnetic annealing, charged-particle beam control (e.g., wiggler magnets for synchrotron light sources), MRI, NMR spectroscopy, and oil and water treatment, among other applications. In the laboratory, permanent magnet flux sources of this type are the most versatile and convenient way of generating variable fields in the range 1 mT to 2 T. They are compact, and need no cooling water or high-current power supply. However, the limitation on the polarization of existing (Table 1) and foreseeable hard magnetic materials to less than ~2 T will mean that permanent magnet flux sources will only be economically justified for fields <3 T. For fields >3 T, superconducting solenoids with cryogenic or cryogen-free refrigeration will be needed. Fields >20 T call for Bitter or hybrid magnets. For most purposes, however, permanent magnet flux sources are destined to displace the iron-yoked electromagnet from the dominant position it has occupied over the past 150 years.

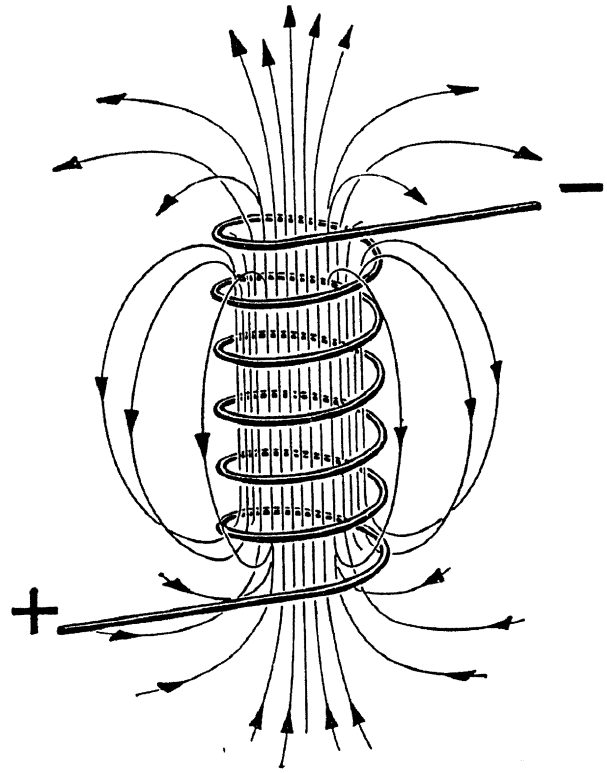


Figure 4. The magnetic field of a coil. (Courtesy of Gerthsen, 1966.)

Electromagnets

In this section, we consider a few basic relationships between current density and magnetic field for a solenoid coil geometry. In the second part, we briefly deal with electromagnetic forces ("Lorentz force") and a simplified description of the resulting stresses.

Figure 4 shows a single-layer coil, wound from a piece of conductor, and the resulting magnetic field as described by the magnetic field lines. Solenoids are coils with many turns and layers, and the current can be described as a current density in a volume element. The law of Biot and Savart gives the field contribution at the center of the coil, i.e., at $r = 0$ and $z = 0$, of a small volume element dV carrying the current density j [as in the literature, j is used here for current density (in units of A/m^2), to be distinguished from polarization J (in units of T) in permanent magnets] in circumferential direction (Fig. 5).

$$\mathbf{H} = \frac{1}{4\pi}\int \frac{\mathbf{r}' \times \mathbf{j}}{r'^3}\, dV \qquad (5)$$

From symmetry consideration it follows that on axis we only have an axial field component, B_z, which we can calculate by integrating over a magnet with inner radius a_1, outer radius a_2, total length $2b$, as shown in Figure 5. From here on, we use the magnetic induction $B = \mu_0 H$ for the magnetic field

$$\mathbf{B} = \frac{\mu_0}{4\pi}\int_0^{2\pi}\int_{a_1}^{a_2} 2\int_0^b \frac{\mathbf{r}' \times \mathbf{j}}{r_3'} r\, dr\, d\varphi\, dz \qquad (6)$$

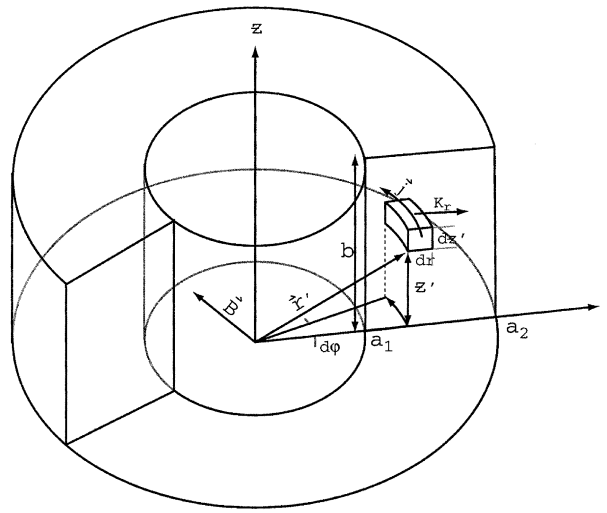


Figure 5. Field contribution of a conductor element.

We obtain for constant current density j over the coil volume the following expression for the magnetic field at the center of the solenoid with $\alpha = a_2/a_1$, and $\beta = b/a_1$.

$$B_z = \mu_0 j \ln \frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{1 + \sqrt{1 + \beta^2}} \quad \text{or} \quad B_z = \mu_0 j a_1 F(\alpha, \beta) \qquad (7)$$

where $F(\alpha,\beta)$ is the so-called field factor that is entirely geometry dependent, i.e., the solenoidal magnet shape can be optimized (Montgomery, 1969). The cross-section of the coil with the overall current density j can be replaced by n turns of wire carrying a current of I occupying only a fraction λ of the total volume. This "space factor," λ, i.e., the ratio between the active section of the winding and its total section, is of great significance, since insulation, reinforcement, and wire shape tend to dilute the wire current density $j_{\text{wire}} = j/\lambda$, and, therefore, the achievable fields. Optimized designs, with the goal of either achieving minimum volume and, therefore minimizing wire cost, or of achieving the highest possible field, will deviate from the above assumption of an overall constant current density and subdivide the winding into regions of different constant current densities to match conductor characteristics with respect to critical current or stress/strain limitations.

Electromagnetic Forces. The design of magnets, especially of high-field magnets, is limited by the Lorentz force, i.e., the stress and strain levels induced by this force, via the coil geometry, on the conductor and the support structure. The precise description of the forces and resulting mechanical loading is extremely complicated and requires a detailed knowledge of the electromagnetic field distribution and the mechanical structure in all three dimensions. Extensive computer programs have been written to solve this issue satisfactorily (e.g., Markiewicz et al., 1999). We present here a simple approach to calculate, in an approximate way, stress levels in a magnet.

The Lorentz force K is a force per volume V (Fig. 5)

$$\frac{d\mathbf{K}}{dV} = \mathbf{j} \times \mathbf{B} \qquad (8)$$

Considering only the z component of the field B_z, a current density in circumferential direction j_t, and no mechanical coupling to other parts of the coil, one obtains a radial force component K_r of

$$dK_r = j_t B_z r \, dr \, dz \, d\varphi \qquad (9)$$

and the hoop stress σ_t of an unsupported current sheet results in

$$\sigma_t = j_t B_z r \qquad (10)$$

For instance, a 10-T solenoid with a current density of 4×10^8 A/m^2 has to support a hoop stress of 200 MPa in its innermost wire layer of 100-mm diameter. Since the magnetic field is proportional to the current density, the Lorentz force increases with the square of the magnetic field. The performance of most magnets and the achievable fields are limited by the materials available to support the induced stress or strain.

Superconducting Magnets

While all metal conductors show a gradual decrease in resistivity with lower temperature, superconductors show zero resistivity very abruptly below a certain temperature, the critical temperature T_c. Another typical property of superconductors is that they have a critical field, B_c, and a critical current density j_c, beyond which superconductivity is destroyed. Superconducting magnets have to be operated within the described B-j-T volume (A-C-E in Fig. 6) since the superconductor transitions into normal resistivity outside this volume. The prevailing conductors are alloys of niobium-titanium and niobium-tin with $T_c s$ of $\sim$12 K and 27 K, respectively (C in Fig. 6). Therefore, the magnets are operated at the temperature of liquid helium (4.2 K) or superfluid helium (1.9 K) (point F). At that temperature the conductor has a lower maximum current density (point D) at zero field, which is again considerably lower in a magnetic field (point G). A magnet is therefore operated along the load line F-G. The new development of high-temperature superconductors (HTS) will extend the useful region toward 100 K; however, at present they are competitive only at low temperatures. Figure 7 compares promising HTS conductors with commercially available low-temperature superconductors.

The generation of magnetic fields through the application of superconductivity can now be considered as a well established technology. Many difficult technical problems have been resolved in the last few years. Thousands of small research magnets are now in routine operation worldwide, producing fields in the range from 5 to 20 T. Both NMR and MRI are the key industrial applications (Killoran, 1997). These applications require extraordinarily high homogeneity of the magnetic field in space and time—typically 10^{-9}—which is achievable solely with superconducting technology. When the magnet is charged to full current, a superconducting switch is closed across the terminals, x, and the power supply is disconnected.

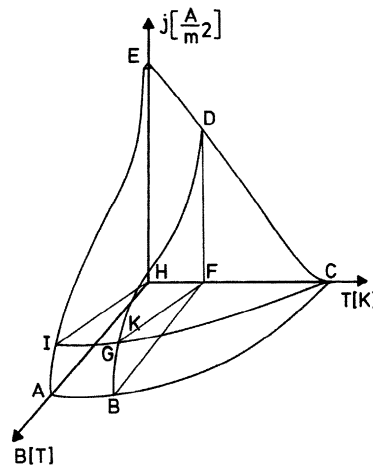


Figure 6. Critical B-j-T volume of a superconductor.

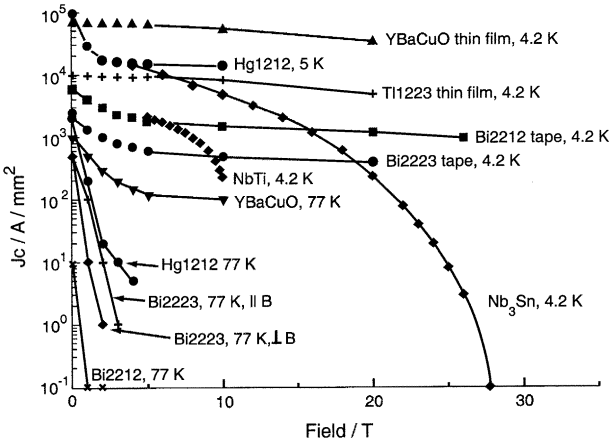


Figure 7. Critical current density within the superconductor of high-temperature superconductors as a function of field for 4.2 K and 77 K. NbTi and Nb_3Sn are plotted for comparison.

The current continues to flow in a closed loop ("persistent mode"), and because of its extremely low resistance—with a decay time of the order of 100,000 years—the magnet is almost a *perpetuum mobile*.

Quench and Training. Superconducting magnets quite often show a performance that is lower than the expectation, which is the field value computed from the short conductor sample characteristics; i.e., the magnets quench prematurely. This is indicated by point K on Figure 6 compared to G. The word "quench" has been coined to describe a process wherein the conductor, starting from a point in a magnet and expanding over the entire volume, passes from the superconducting into the normal state. The stored magnetic energy is dissipated as thermal energy within the coil volume. The heating is not uniform and "hot spots" are generated. The temperature peaks at the origin of the quench and can even result in the melting of the conductor. Modern designs, especially for magnets with high energy content, provide a quench-protection system and even heaters to assure uniform energy dissipation and temperature. Many magnets show a progressive improvement up to a certain level with repeated quenching. This effect is called "training." There is common agreement today that the problem is created by some sudden energy release as the current, and with it the mechanical loading through the Lorentz force, is ramped up. This may originate from some weak spots in the winding structure, cracking of the impregnation under the impact of the induced strain, and small rearranging of the conductor with friction. Because of the extremely small heat capacity of the materials at low temperatures, small energy releases create substantial temperature increases that bring the conductor temperature beyond the critical temperature, T_c. Heat conduction within the winding pack is poor and slow; therefore, the assumption of adiabatic heating is justified. The results necessitate that disturbances must be limited to extremely low values of the order of microjoules per cubic millimeter, corresponding to a maximum sudden wire movement of only a few micrometers. A sound mechanical structure, a precise winding pattern, crack-free epoxy impregnation, a design without slip planes and shear force releases, and a mindful attention to the many mechanical details, have proven to prevent training.

Protection. During a quench, a magnet may be exposed to excessive voltages or temperatures, and the magnet has to be protected against any damage. There are active and passive techniques for doing this. The active technique involves the discharge of the magnet energy into an external dump resistor. In the interest of maximum energy extraction, a small time constant $\tau = L/R_e$ (L is the inductance of the magnet, R_e the external resistor) is preferable. The discharge voltage across the terminals, $V_e = I_m R_e$ (I_m is the magnet current), however, increases and may damage components by arcing. The magnet designer's job is to find a compromise between maximum quench temperature and discharge voltage. This involves the appropriate selection of the operating current (defined by the energy content; $E = 0.5LI_m^2$), the conductor cross-section, and the inductance. A significant feature is a completely reliable quench detection and switch operation system.

Passive techniques are, per definition, reliable. The disadvantage, however, is that all of the energy is dissipated into the magnet volume and the coolant, i.e., liquid helium. The preferred solution is subdivision of the winding into a sufficiently large number of sections that are shunted by a resistor. The resistors can also be used to heat the windings to initiate a distributed quench for a more uniform temperature distribution. Another possibility exists in integrating shorted coupled resistive windings that take up a fraction of the energy. An optimized design will divide the magnet into many subcoils, grade the conductor to suit the local field, construct the coil reinforcement as a wire wound and short circuited overbending for quenchback and heating, and combine it with a quench detection system that dissipates the coil energy in quench heaters. Computer codes have been written that predict quench propagation and coil currents quite reliably by including ac losses (Eyssa et al., 1997). The passive protection has the disadvantage that there is also an energy loss, and therefore increased helium losses, when the magnet is energized or the current setting is altered.

Stability and Losses Due to Changing Fields. The critical-state model describes the penetration of a magnetic field into a superconductor. Because of the superconductor's zero resistivity, screening currents are induced. If the critical current density is surpassed, the screening currents decay resistively to the critical current, allowing the magnetic field to penetrate. All regions of a superconductor are, therefore, carrying either currents at the level of the critical current or no current at all. At full penetration, the current reaches the critical value, I_c. This state may become unstable because the critical current density falls with increasing temperature, and flux motion in the superconductor generates heat. The so-called "adiabatic stability criterion" can be derived from this, with the result that sufficiently thin conductors are stable against flux jumps. In a composite conductor, and above its critical current, the excess current is carried by the stabilizer. A voltage drop is set up which is determined by the stabilizer

resistivity. Copper is usually selected as matrix material as it provides good electrical and thermal conductivity and is essential to protect the conductor against burnout in case of a quench. However, a copper matrix retains the handicap that the filaments are coupled in a changing field via the resistance of the matrix. The solution to this problem consists of twisting the conductors, so that the induced currents cancel over one twist pitch. Twisting is effective against external fields; however, it is not effective against the self-field, induced by the transport current within the conductor. This then leads to the "transport current stability criterion," predicting that wires up to diameters of ~0.5 mm will be stable against flux jumps for the adiabatic case. Further improvements in the theory of stability have been achieved by including the time dependence of heat, current, and magnetic flux, resulting in the so-called "dynamic stability conditions." Obviously, heat conduction and surface cooling are most important for dynamic stability. Experience has shown, nevertheless, that filamentary conductors perform better than theory predicts. As a consequence, modern superconductors are the result of an ongoing development effort, and are tailored to their applications. Figure 8 shows, as an example of the highly advanced wire technology, a bronze route conductor developed especially for high-field NMR magnets.

A detailed knowledge of the induced losses in a changing magnetic field is important for determining operating conditions and the load for the cryogenic system. In superconductors, as in all conductors, changing magnetic fields induce voltages that generate a current flow through the resistance of the current loop. The difference to resistive conductors arises from the fact that the changing magnetic field generates a varying magnetic flux in the filaments. This variable flux results in induced voltages that act on the shielding currents. Heat is dissipated as described above, i.e., the loss mechanism is of a resistive nature. The resistance is highly nonlinear and the loss per cycle is independent of the cycle time. These losses are called hysteresis losses. Clearly, the induced currents, and with it the losses, depend to a high degree on the design of the conductor, the resistivity of the matrix, the resistance between filaments and the matrix ("coupling losses"), the direction of the pulsed magnetic field, and the twist pitch. In addition, we must deal with self-field losses if the transport current changes. Collectively, these losses are known as "ac losses," and their calculation is highly complicated. Many industrial applications (e.g., particle accelerator magnets, superconducting motors and generators, levitated trains, or energy-storage systems) involve changing magnetic fields. In general, hysteresis losses can be reduced by making the filaments as fine as is feasible. Coupling losses can be minimized by shortening the twist pitch, but only if the changing field is perpendicular to the conductor. Self-field losses can be controlled only by reducing the wire diameter.

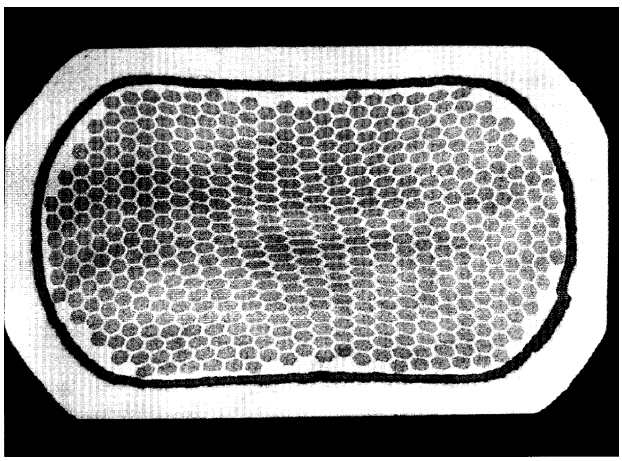


Figure 8. Modern large cross-section (3.17 × 1.97 mm^2) Nb_3Sn conductor with outer stabilizer (25% Cu) and Ta barrier. The conductor contains 123,000 filaments of 4-μm diameter. The critical current density at 20 T and 4.2 K is ~100 A/mm^2, and 200 A/mm^2 at 1.8 K. (Courtesy of Vacuumschmelze, Hanau, Germany.)

Resistive Magnets

Resistive magnets, unlike superconducting magnets, have no constraints, such as critical current density or temperature, imposed on their achievable performance. Resistive magnets require electrical power and, therefore, efficient cooling, but are not limited by any physical effect as superconductors—i.e., in principle it is possible to generate an infinitely high magnetic field. We will see later that power requirements become excessive very quickly, therefore, the generation of very high magnetic fields is only a question of economics.

Resistive magnets are optimized to generate the highest magnetic field possible with the available power. We will present the basic relationships between field, power, and different current density distributions to define the criteria for an optimum design.

Power-Field Relations. A small conductor element dV as shown in Figure 5 with resistivity ρ and current density j will dissipate the following amount of power

$$dW = j^2 \rho \lambda \, dV \tag{11}$$

If we assume a radial dependence of the current density, $j(r)$, but constant resistivity, ρ, and space factor λ, we obtain for a rectangular coil cross-section

$$W = 4\pi b \rho \lambda \int_{a_1}^{a_2} j(r)^2 r \, dr \tag{12}$$

From Equation 6, the same assumptions yield the following:

$$B_z = \mu_0 \int_{a_1}^{a_2} \frac{b}{\sqrt{b^2 + r^2}} j(r) dr \tag{13}$$

Equations 12 and 13 can be combined to the following general form:

$$B_z = \mu_0 G(\alpha, \beta) \sqrt{\frac{\lambda W}{\rho a_1}} \tag{14}$$

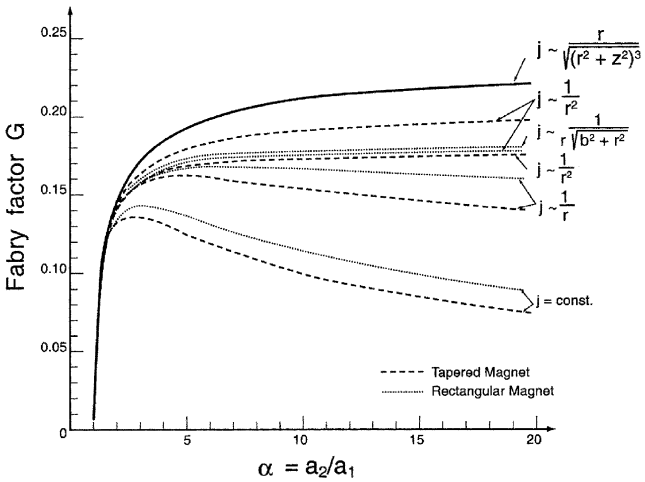


Figure 9. The Fabry factor for different radial current densities. β is optimized for each α value.

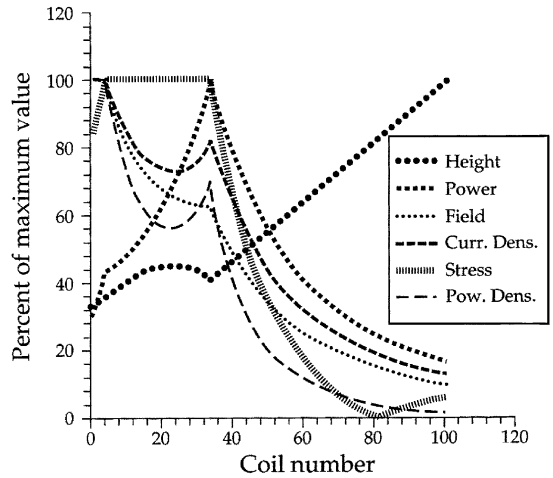


Figure 10. Optimization of a resistive magnet with the constraints of maximum power density and hoop stress.

where $G(\alpha,\beta)$ is the so-called Fabry factor, which is a geometry factor, and B_z is the magnetic field generated at the origin, i.e., in the midplane of the solenoid at $r = z = 0$. Since Equation 14 relates the generated field to the consumed power, it describes the efficiency of the chosen radial current density distribution and solenoid geometry. We can now compare the G-factor of different current density distributions and geometry. Figure 9 shows that, for constant current density and for $j \sim 1/r$, G has a maximum, but that for the other current density distributions, the efficiency can be increased by making the magnet bigger and bigger. Of special interest is the so-called Kelvin distribution (top line in Fig. 9), as it shows the highest efficiency. It is plotted here for a solution that allows the current density to change radially and axially. It is based on the concept that each volume element of the magnet should produce the same amount of field per watt of power input. Practical solutions approximate the optimum value quite closely by subdividing the magnet into several coils and properly selecting the power distribution.

The relationships shown here hold true for magnets with low power levels. High-field magnets have many additional constraints. The power levels are so extremely high that evacuation of the dissipated power through cooling, i.e., the heat flux, is limited by the available cooling surface. Additional cooling requires increased space, no longer available for current conduction, i.e., the space factor λ becomes a function of the power density. The stress levels are so intense that the use of copper alloys is necessitated, i.e., the resistivity ρ also becomes a function of the power level. Additionally, the required reinforcement further increases the space factor. The equations become so complicated that numerical methods are required to solve them. Figure 10 displays the result of a calculation made to illustrate the various constraints. The current density was illustrated only to vary radially. The magnet was split into 100 coils to obtain a quasi-continuous picture. Three regions can be distinguished: (1) the outer region where the optimization has no constraints and finds $j \sim 1/r^2$ and $b = r$, which is the most efficient geometry and power distribution (see Fig. 9); (2) the center region, where the current density is reduced because of the imposed hoop stress limitation; and (3) the inner region where the current density is reduced because the power density is limited for reasons of insufficient cooling surface. An optimum design will attempt to meet both limitations simultaneously. Due to these constraints, resistive magnets become rather inefficient with higher fields, i.e., their power requirements increase exponentially with field.

As result of a newly developed technology, ("Florida Bitter Magnet"), resistive magnets now generate up to 33 T in a 32-mm room temperature bore (Eyssa et al., 1998). These high-field magnets have a power dissipation of up to 20 MW; therefore, efficient and reliable cooling is a prerequisite. Due to the high power densities, the coolant has to be in direct contact with the conductor. Therefore, deionized water of high electrical resistivity is used. Efficient cooling requires high water velocities and a highly turbulent flow to reduce the laminar boundary layer and the temperature drop across this layer as much as possible. Typical values are 20 m/s for the water velocity as upper limit to avoid cavitation. A laminar layer of about only 5-μm results, which accounts for a temperature gradient of up to 40 K. The roughness and its structure determine heat transfer and friction factor, and its optimization is the aim of extended numerical efforts.

Hybrid Magnets. Hybrid magnets combine the advantages of superconducting and resistive magnets. They consist of an outer superconducting coil that provides the background or booster field for the resistive insert magnet. The outsert requires only negligible power compared to a resistive magnet, and power savings of 10 to 20 MW can be realized, depending on bore diameter and field level. Because of the necessary large bore, however, it represents a major capital investment. For the insert, normal, resistive conductors are utilized, as their current-carrying capability is unlimited in comparison to a superconductor. The design of hybrid magnets is a challenging task. The outer magnet has to generate a high field in a large bore

with the consequence of high hoop stresses (as follows from Equation 10). The insert has to withstand not only its own Lorentz forces but also those from the field contribution of the outsert. For economic reasons, the bore size of the outsert is kept as small as possible; therefore, the insert must dissipate its power in a rather small volume. In the case of a misalignment of the magnetic centers of the two systems, forces are generated that are transmitted through the cryostat of the superconducting magnet. In case of a partial burn-out or short-circuit of one of the magnets, these forces can reach the alarming magnitude of several MN. Because of the magnetic coupling, the outsert must also support induced currents arising from sudden field changes from the insert, for instance, in case of an emergency switch-off. Stability requirements and the Lorentz forces lead to a special conductor design for the outsert, the so-called cable-in-conduit conductor, where the conductor is in direct contact with superfluid helium. The stainless steel shell handles the mechanical forces.

There are several hybrid magnets in service worldwide. As of 1999, the highest field (44 T) was generated in the U.S. at the National High Magnetic Field Laboratory (NHMFL), followed by the magnet laboratory in Tsukuba, Japan (37 T). The NHMFL is working on a new insert design that will push the limit even higher, with a goal of generating 50 T in a 32-mm bore.

Pulse Magnets

The generation of pulsed magnetic fields represents an economic avenue to high fields at a small fraction of the cost of continuous magnets. Pulsed magnets are the only way to produce fields above the level of continuous magnets. Disadvantages are the induced voltages and eddy currents in the sample through the changing field. Pulse lengths are typically in the micro- to millisecond range. The use of pulsed fields for research purposes is, therefore, limited to systems where the above effects can be avoided, for instance, in samples of low conductivity. Modern electronics offer fast acquisition time and high resolution.

Devices to generate pulsed fields consist of an energy-storage system that is charged slowly and then discharged rapidly on the magnet. This may be a capacitor bank (typically in the range from 1 to 10 MJ) or a generator in which the energy is stored in the rotating mass of the winding, sometimes complimented by a flywheel. Generators can provide stored energies in the 100-MJ to 3-GJ range. There also exists the option of procuring energy pulses out of the power mains, if the locally available power grid is sufficiently stable.

Discharges from capacitor banks are uncontrolled in the sense that the resulting pulse form is determined by the circuit, i.e., a damped sine wave. Its pulse width is set by the inductance L of the magnet and the capacitance C of the bank, and the damping by the coil and circuit resistance. In first approximation, i.e., for negligible damping, we can write the following equation for the peak current (where V_0 is the voltage of the capacitor bank)

$$I_{\max} = V_0\sqrt{\frac{C}{L}} \tag{15}$$

Similarly, the following equations can be written for the peak power

$$N = V_0^2\sqrt{\frac{C}{L}} \tag{16}$$

and for the pulse-rise time

$$t_0 = \frac{\pi}{2}\sqrt{LC} \tag{17}$$

For the generation of high fields, a high peak current, and equally importantly a high power transfer, from the bank to the coil is needed. This requires a high-voltage capacitor bank and a low-inductance coil. For example, a 10-kV, 1-MJ bank would deliver in excess of 50 kA and 500 MW into an 80-T magnet (Li et al., 1998). Assuming an inner coil diameter of 10 mm, the stress on the conductor will be 2 GPa (Equation 10), i.e., in the range of the strongest materials available today. Often a "crowbar" is added to the discharge circuit to increase the pulse length. At peak current, the capacitor bank is disconnected and replaced by a resistor R, so that the current decays in the LR circuit with a time constant τ of

$$\tau = \frac{L}{R+r} \tag{18}$$

where r is the (time dependent) resistance of the magnet. The total pulse time is then $t \approx t_0 + 3\tau$.

Generators and power supplies fed from the power mains require a unit that controls the energy flow to the magnet and also rectifies the current. The control unit can be used to achieve various pulse forms. Figure 11 gives examples of the pulse shapes that can be generated with the unique 60-T magnet at the user facility of the NHMFL at Los Alamos National Laboratory (LANL).

The upper limit for the above-described technique is considered to be 100 T. There are no known materials that can withstand the Lorentz forces beyond this field level, and magnets inevitably become destructive. To

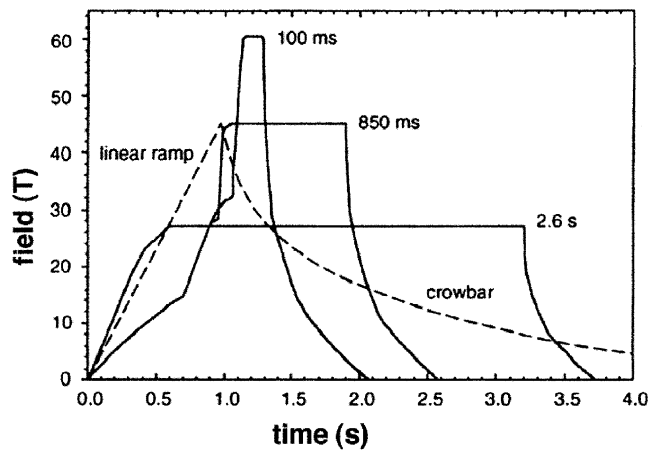


Figure 11. Some of the pulse forms obtainable with a controlled discharge at NHMFL user facility at LANL.

generate even higher fields, two concepts are applied. The first concept is delivering power into a coil faster than the time the coil takes to explode. Fields up to 300 T during microseconds can be created this way by the discharge of a low-inductance capacitor bank into a single turn coil of a few millimeters diameter. Although the coil explodes, because of the symmetry of the explosion, cryostat and sample remain intact. The second concept is to compress a large volume magnetic field into a small volume. Fields up to 500 T can be achieved through electromagnetic flux compression, and well above 1000 T through flux compression with explosives. In both cases, the peak field is achieved in a few microseconds, and the sample is destroyed at this moment.

MAGNET LABORATORIES FOR USERS

It may have become clear from the above that design, construction and operation of high-field magnets is a very challenging and also very costly task, requiring considerable efforts in manpower and installations. It is therefore obvious that a few specialized laboratories of significant size have developed with the goal of serving a broad user community. In the following we give a short overview of the most important laboratories. A detailed review can be found in Herlach and Perenboom (1995).

Continuous Magnetic Fields

It is interesting to realize that the number of laboratories where magnetic fields are used as a research tool has grown considerably over the last several years, accompanied by a more than proportional growth of publications in this area. The strong trend toward higher fields, made possible by new investments (Tallahassee, Tsukuba, Grenoble, and Nijmegen), and the development and evolution of new technologies and materials, will continue. A large and growing user community takes advantage of these central facilities, which are well-equipped with magnets in the 20- to 45-T range, along with the necessary scientific instrumentation (Crow et al., 1996). At present, the development goal is 50 T as the highest continuous field.

Superconducting magnets are now available up to 20 T, and many laboratories employ such a system. We list here only laboratories which can offer continuous fields above 25 T with resistive and hybrid magnets. These are: High Magnetic Field Laboratory, Grenoble, France; Nijmegen High Field Magnet Laboratory, Nijmegen, The Netherlands; High Field Laboratory for Superconducting Materials, Sendai, Japan; National High Magnetic Field Laboratory, Tallahassee, Florida, U.S.A.; and Tsukuba Magnet Laboratories, Tsukuba, Japan.

Pulsed Magnetic Fields

Pulsed magnetic fields represent an economical access to high fields in the 50- to 80-T range. Worldwide, many laboratories provide assistance and support a user program. Experimentation has become easier through the development of new techniques, such as digital data recording and miniaturization. Ultralow temperatures and extreme high pressures have been combined successfully with pulsed fields (Herlach, 1985). The most remarkable development is a 100-T system that will generate pulses in the 10- to 20-ms range, a factor of over 1000 above existing systems (Baca et al., 1999).

There are ~30 laboratories worldwide that do experimental work in pulsed magnetic fields, and which provide assistance to users. We list here only a subgroup, selected from Herlach and Perenboom (1995), which provide fields above 60 T. These are: Fachbereich Physik, Humboldt Universität, Berlin, Germany; H.H. Wills Physics Laboratory, University of Bristol, Bristol, U.K.; Katholieke Universiteit Leuven, Leuven, Belgium; National High Magnetic Field Laboratory, Los Alamos, New Mexico, U.S.A.; Research Center for Extreme Materials, Osaka University, Osaka, Japan; Princeton Plasma Physics Laboratory, Princeton University, Princeton, New Jersey, U.S.A.; National Pulsed Magnet Laboratory, University of New South Wales, Sydney, Australia; Megagauss Laboratory, Institute for Solid State Physics, Tokyo, Japan; Service National des Champs Magnétiques Pulsés, CNRS, Toulouse, France; and Tsukuba Magnet Laboratories, NRIM, Tsukuba, Japan.

A few laboratories provide fields above 100 T. These are: Fachbereich Physik, Humboldt Universität, Berlin, Germany; National High Magnetic Field Laboratory, Los Alamos, New Mexico, U.S.A.; Scientific Research Institute of Experimental Physics, VNIIEF, Sarov, Russia; Pulsed Magnetic Field Installation, Russian Academy of Science, St. Petersburg, Russia; and Megagauss Laboratory, Institute for Solid State Physics, Tokyo, Japan.

MEASUREMENTS OF MAGNETIC FIELDS

The knowledge about the strength and direction of a magnetic field is essential for the following three main purposes.

1. Calibration of a magnet as function of another parameter, typically the magnet current, so that it can be used for its application. The result also tells the magnet engineer how much the constructed magnet corresponds to its design values.
2. Determination of the magnetic environment of a sample to be investigated. Magnetic fields are used as a tool to measure the dependence of some parameters as a function of the applied magnetic field. Therefore, the investigator has to verify, through measurements, the properties of the desired field, e.g., strength, direction, homogeneity, and stability. Equally important for reference measurements is the knowledge of the zero field environment when the magnet is switched off. Magnetization of ferromagnetic parts, for instance, as housing or support structure, can produce noticeable low-field contributions.
3. Fringe fields of magnets have an impact on electronic devices and machinery. The knowledge of their existence is required to initiate and determine the necessary amount of countermeasures, such as

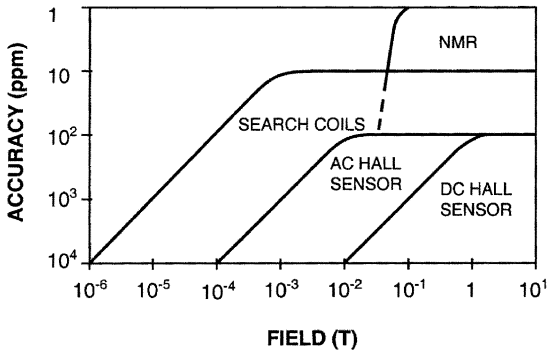


Figure 12. The main measurement methods for magnetic fields and their accuracy and field range. (After Henrichsen, 1992.)

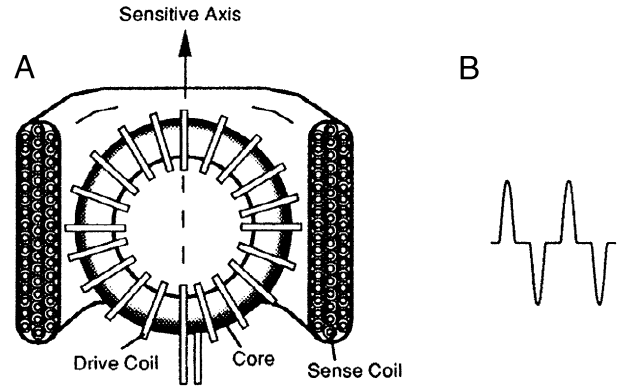


Figure 13. (**A**) Magnetometer with a toroidal core and (**B**) output pulses. (After Fraden, 1993.)

shielding or orientation with respect to the field lines. Fringe fields of high-field magnets can be important, especially with magnets with large bore diameters. Safety measures must be considered for health reasons (pacemakers, for instance), and the force on ferromagnetic parts, like chairs, racks, tools, or gas bottles. The space around a magnet, therefore, has to be cleared carefully of any apparatus and must be cordoned off as a restricted area for the protection of personnel.

The following describes a selection of important measuring devices of magnetic fields. We start the discussion with low-field devices: the compass and the flux-gate magnetometer. We then concentrate on medium-to-high-field measuring techniques: search coils, Hall sensors, and NMR devices. The Hall sensor is the singularly most important, and the most commonly used, device in the measurement of magnetic fields; as such we provide a more detailed description of a new development, the silicon Hall sensor. Figure 12 is a comparison of the most important measurement methods in view of their accuracy and effective useful field range (Henrichsen, 1992).

A large variety of instruments are commercially available for the measurement of magnetic fields.

Low-Field Sensors

Fringe Field Sensors. The existence of low magnetic fields, such as fringe fields of magnets or the fields of magnetized parts, can easily be demonstrated by a simple device: the magnetic compass. The magnetic compass was invented in China about 2500 B.C. and played a role in navigation as early as the 10th century A.D. For a simple demonstration of fringe fields, a screwdriver suspended on a thin thread works well. This can also serve as an indicator of the direction of the field lines and be used for a visual field mapping.

Another simple method to detect low magnetic fields is the monitor of a computer. Black and white cathode ray tube (CRT) monitors change their pattern at ~1 mT; color CRT monitors change their colors at about half that value and therefore, are also a very simple way to check for fringe fields.

Flux-Gate Magnetometer. The flux-gate magnetometer can be utilized for accurate measurements of weak magnetic fields. It is based on a ferromagnetic core with a rectangular hysteresis curve on which detection and excitation coils are wound. Whenever magnetic flux lines are drawn inside the ferromagnetic core, a pulse is induced in the detection coil. A pulse of opposite sign occurs when flux lines are pulled out. The amplitudes of these pulses are directly proportional to the flux intensity; their polarity indicates the direction of the flux lines. The alternate excitation current drives the core into and out of saturation. Figure 13 shows a magnetometer with a toroidal core. It has the advantage that the excitation flux cannot induce current in the sense coil. Flux-gate magnetometers offer linear measurement of the field and its direction, and are especially well suited for the detection of weak stray fields around magnets. Sensitivity can be as low as 20 pT. Commercial devices offer ranges of ~1 mT with a resolution of 1 nT.

High-Field Sensors

Hall Effect Sensors. The Hall effect is a galvanomagnetic effect that occurs by the interrelationship between the electrical currents flowing in a material that is being subjected to a magnetic field (see DEEP-LEVEL TRANSIENT SPECTROSCOPY). It can best be exploited in semiconductor materials in which a high carrier mobility and a low free carrier concentration is achievable. The force F on a free charge carrier with charge e in a semiconductor is driven by the electric field F_E and deflected by the Lorentz force F_L, which acts perpendicular to the direction of the motion and the magnetic flux density

$$\mathbf{F} = \mathbf{F}_E + \mathbf{F}_L = e\mathbf{E} + e[\mathbf{v} \times \mathbf{B}] \tag{19}$$

If a plate-shape piece of semiconductor material in the xy plane carries a current I in the x direction, then an applied flux density B in the z direction produces a transverse electric field in the y direction, the Hall field (Fig. 14).

The integral of this Hall field over the plate can be measured at its outer edges as the Hall voltage, V_H, and

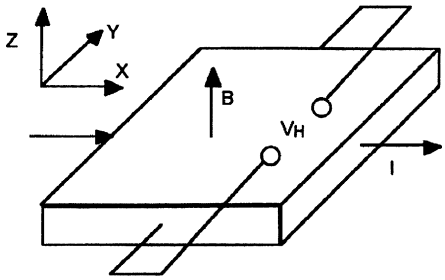


Figure 14. The Lorentz force acting on moving charge carriers in a semiconductor plate causes them to build up an electric field which can be measured by the Hall voltage.

is approximately proportional to the product of the flux density B and the current I

$$V_H = S_H IB \tag{20}$$

Magnetometers/Compound Semiconductor Sensors. A typical Hall effect magnetometer, often called a gaussmeter, incorporates several circuits and controls: a current supply for the sensor, an amplifier with sufficient gain to raise the level of the sensor output to drive an analog/digital converter, zero and calibration adjustments, and analog and/or digital display and voltage output, and either analog or digital signal processing hardware, as well as software to compensate for probe-temperature effects, nonlinearities, and voltage offsets. To date, the most commonly used Hall probes have been based on Hall generators made of compound semiconductors such as InSb and especially InAs. Such generators have a sensitivity S_H (as defined by Equation 20) ranging from 50 mV/AT to 3 V/AT, and enable the gaussmeter to offer full-scale ranges from 30 μT to 30 T, with an accuracy in the 0.1% to 1% range and a precision as high as 10 ppm full scale. Several versions of these Hall elements have been used to measure magnetic fields as high as ~34 T and at temperatures as low as 1.5 K (Rubin et al., 1975, 1986).

Theoretically, a high carrier mobility is crucial for achieving high sensitivity, low offset, and low noise. However, the application of the compound semiconductor Hall elements also implies some drawbacks, such as long-term instability and high-temperature cross-sensitivity. To overcome these difficulties, nowadays high-precision magnetometers are increasingly based on silicon Hall generators.

Silicon Hall Sensors. Although silicon is not a high-carrier-mobility material, it offers many other advantages that make up for this drawback. The doping is usually n-type, with a density around $10^{15}/cm^3$, so that at room temperature the device can be considered as strongly extrinsic. Large-series fabrication of these devices and the application of highly sophisticated and mature silicon technology result in narrow tolerances for the parameters. Moreover, the parasitic effects in these devices are well understood and can be described by simple equations. These characteristics allow efficient means to develop structural and analog electronic compensation of all parasitic effects. A typical value for the proportionality factor (equivalent to sensitivity) S_H, in a modern silicon device is 400 V/AT.

The Integrated Vertical Hall Sensor. Whereas early devices were made as bulk or thin-film discrete components, modern Hall sensors are made using highly advanced semiconductor integration technology, making large numbers of high-quality devices available at low cost. In addition, the sensors can be manufactured as buried devices featuring low noise and very good long-term stability. Their extremely small dimensions make them especially suitable for measurements with high spatial resolution, even in highly inhomogeneous fields.

Besides integrated Hall plates with conventional geometry, the above-described technology also made it possible to implement a sensor measuring the magnetic flux density parallel to the chip surface. In the vertical Hall device (Popovic, 1995), the general plate shape is chosen in such a way that all electrical contacts become available at the surface of the chip. Its geometry evolves from the conventional shape, applying a conformal mapping method where all angles are conserved along with the galvanomagnetic effects mentioned before (Fig. 15). Such devices are adapted to fabrication by integrated technology in an almost perfect way, since the main sources for noise and offset can be reduced to a minimum, yielding highly accurate magnetic sensors. These structures have also been realized to measure two or three components of the magnetic field synchronously in the same spot (Schott and Popovic, 1999), making them suitable for highly demanding tasks like scanning or controlling fields in rather complex magnetic systems.

Even in a modern vertical Hall magnetic sensor, the proportionality between the electrical output signal and the field applied on the device is not ideal. This is due to physical effects within the material of the active zone (charge carrier transport phenomena) as well as the structural composition of the device (depletion zones, surface effects) and the tolerances of the fabrication process itself (mask misalignment, gradients in doping profile). When working at high accuracy, all of these effects have to be taken into account. This can be achieved by applying two different compensation strategies: internal and external. Internal compensation eliminates the influences of offset and of voltage-dependent modulation of the active zone. It is based on mutual cancellation of parasitic effects by adding them with opposite sign.

External compensation eliminates the remaining quadratic temperature cross-sensitivity and the problem of nonlinear behavior in stronger fields. The Hall device follows simple equations with these parameters, so that the analog electronic compensation function can easily be synthesized. Operating the Hall devices in constant

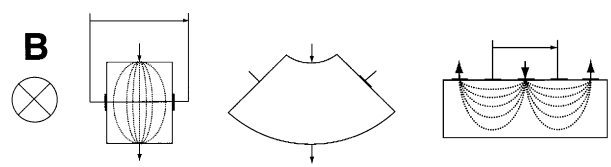


Figure 15. Derivation of the vertical Hall device by conformal mapping of Figure 14.

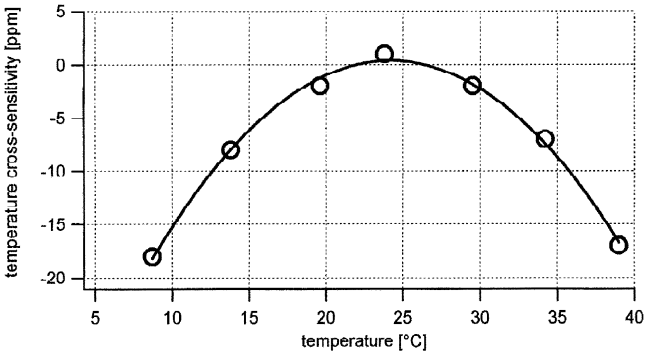


Figure 16. Dependence of sensitivity on temperature. A typical vertical Hall device varies $<5 \times 10^{-5}$ within ± 10 K around room temperature.

current mode allows for implementing all external compensation through bias-current modulation.

Temperature Cross-sensitivity. The Hall factor for electrons in n-type silicon around room temperature increases $\sim 0.1\%$/K. The device's input resistance also rises linearly with temperature. Profiting from this fact, the Hall device itself can be used as a temperature sensor. A simple resistor, put in parallel with the device's input resistance, compensates perfectly for first-order temperature dependence. The less critical remaining parabolic drop of sensitivity with temperature is eliminated by adding a corresponding supplementary current to the biasing current. As a result, sensitivity stays constant over a wide temperature range (Fig. 16).

Offset. Offset in integrated sensors is mainly due to doping nonuniformity and mask misalignment during the fabrication process. According to the Hall device circuit model (Popovic, 1995), offset can be put down to an asymmetry of the resistances between the current contacts and the Hall contacts. Therefore, it is directly correlated to the material resistivity, causing it to rise with temperature and, due to the magnetoresistance effect, also with magnetic field. Since it cannot be distinguished from the output signal resulting from a dc or slowly varying magnetic field, it has to be compensated internally. A first, coarse compensation is provided by the probe's structure (Fig. 17). Two

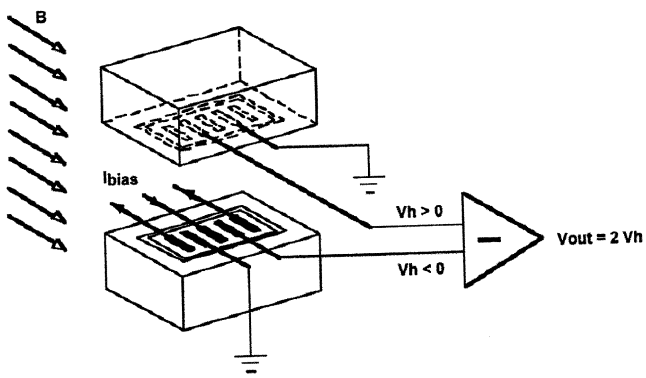


Figure 17. Two Hall devices, oriented in the field so that the Hall voltages are opposed, allow for direct compensation of offset and active zone modulation.

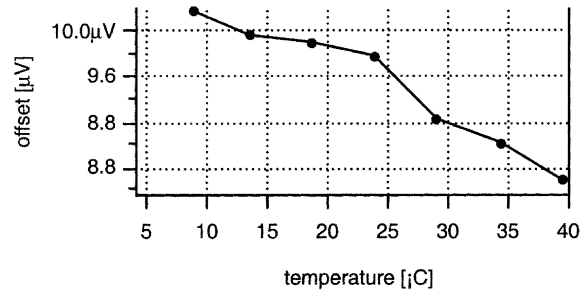


Figure 18. Dependence of the offset on temperature. The equivalent offset field in the specified temperature range from 15 to 35°C is <10 µT.

Hall devices, preferably neighboring elements from the wafer, are subjected to the same parameters during the fabrication process, and therefore, have similar offset. When mounted in such a way that they see an applied field in opposite directions, the value of the measured field can be extracted by subtracting their Hall voltages. This value contains only a very small residual offset, which is the difference in offset between both devices. Compensation of this residual offset makes use of the good correlation between offset drift and input voltage drift with temperature. It consists of subtracting a fraction of the input voltage from the output Hall voltage by electronic means. After the application of these methods, the remaining offset and its drift with temperature are limited to a minimum (Fig. 18).

Nonlinearity. Nonlinearity of Hall devices is described by the variation of sensitivity with the magnetic field H

$$\mathrm{NL} = (S_\mathrm{H} - S_{\mathrm{H}_0})/S_{\mathrm{H}_0}, \qquad \text{where} \quad S_\mathrm{H} = \Delta V_\mathrm{H}/\Delta B \quad (21)$$

Three different sources of nonlinearity can be distinguished. Two of them are technology-independent and described as material nonlinearity and geometry-related nonlinearity. These two depend on the same physical parameters in a similar way, so that they can be used for mutual cancellation in an adapted design. Residual nonlinearity is compensated by modulating the bias current through the Hall elements in a polynomial manner with the applied field, so that field transducers with linearity as good as 0.01% in fields up to 2 T can be produced. The third effect is of specific interest in buried devices like the vertical Hall device. In order to decrease leakage current and border effects, the active zone of the vertical Hall device is surrounded by a depletion layer, formed by a reverse-polarized pn-diode. Its thickness depends on the potential difference between the diode's p and n contacts. This causes the active volume to change in shape and the device to change in sensitivity with a rising Hall or input voltage, just as in a junction-field-effect transistor. However, these effects are significantly reduced when appropriate means for biasing are applied (Schott and Popovic, 1997) or by using two devices seeing the magnetic field in opposite direction.

Magnetoresistive Sensors. There are several materials that change their resistivity in a magnetic field. At low

temperatures, some materials like bismuth change their resistance by a factor of 10^6. The effect is caused by the Lorentz force, which forces the electrons to move on curved paths between collisions and thereby increases the resistivity. In contrast to the Hall effect sensor, the magnetoresistor, when made as a III-V semiconductor (InSb) can be divided into small, long connected strips. This results in a high initial resistance of 50 to 1000 Ω. Only small currents are needed, with the additional advantage that it is only a two-terminal device. Magnetoresistors are inexpensive and have wide industrial applications. They have proven to be useful for scientific measurements; with temperature stabilization an accuracy of 0.1% can be achieved (Weld and Mace, 1970; Brouwer et al., 1992).

Search Coils. Search coils are flux meters that consist of a pick-up coil and an integrator. The pick-up or search coil is exposed to a changing magnetic field. The induced voltage V is proportional to the change of the flux Φ threading the coil, i.e., its number of turns, N, and the coil area, A.

$$V = -NA\frac{dB}{dt} \tag{22}$$

With an NA of 1 m^2, a field change of 1 T will generate 1 V. The coil can be inserted in a field, i.e., moved from a place with $B = 0$ to the place where the field is to be measured, or vice versa. The difference between the two measurements will give a good indication of the accuracy of the chosen setup. In the case of a solenoid, a search coil is moved from a shielded space ($B = 0$) outside the magnet to the center, and the voltage is measured via an integrator. The coil is then moved back into the shielded space, and the voltage is measured again. In a perfect setup, both measurements should give the same result. Obvious requirements are that the induced voltages in the leads be negligible, and that the mechanical positioning of the coil and its angle with the field at the magnet center be precise. There are alternate solutions to the described method, which are to place the coil in the magnet center and ramp the magnet, or to flip the coil around its axis perpendicular to the magnetic field vector.

The accuracy of the search coil method depends on the knowledge of NA and the characteristics of the integrator. Commercially available fluxmeters incorporate state-of-the-art electronic integrators with drifts as low as ±1 μV/min. In addition to dc field measurement, they can measure ac fields at frequencies up to 100 kHz; the accuracy is several percent. For high-accuracy measurements, special attention must be paid to temperature control of the integrator, shielding, low-frequency noise, thermal voltages, and geometry changes of the search coil due to insertion into cryogenic fluids. Very high accuracies may be obtained with differential measurement techniques wherein a pair of search coils is connected in opposition with one coil moving and the other one fixed. Magnet current fluctuations can be eliminated this way. A large variety of search coil geometries and complex harmonic coil systems have been developed. For instance, the flux ball (Brown and Sweer, 1995) makes it possible to perform point measurements in inhomogeneous fields.

A search coil may also be rotated at some frequency, ω, and an ac voltage will develop. The value of B can then be determined by measuring either the peak-to-peak voltage, rms or with a phase-sensitive detector. The complete system is called a rotating-coil gaussmeter and is commercially available.

Search coils or pick-up coils are the ideal way to measure pulsed fields. Because of the high fields and the short times, relatively high voltages are generated and attention has to be given to insulation. For integration, simple passive RC integrators matching the cable impedance can be used.

Nuclear Magnetic Resonance. Nuclear magnetic resonance is a technique that can measure the magnitude of magnetic fields to high precision, down to parts per billion (NUCLEAR MAGNETIC RESONANCE IMAGING). In commercial NMR systems an important subsystem is the field frequency NMR lock. This device uses NMR signals to detect and control the drift of the magnetic field, measure and eliminate variations in the magnetic field intensity across the sample, and measure the average field intensity at the sample position.

NMR measures the motion of nuclear magnetization in a magnetic field. The nuclear magnetization comes from the nuclei (i.e., hydrogen, phosphorus, sodium, and silicon, to name a few) that possess a magnetic moment μ ($\mu = \gamma I$ with γ gyromagnetic ratio, I nuclear angular momentum.) When nuclei having a magnetic moment are placed in a magnetic field, H, the nuclei align parallel to the field. The collection of aligned nuclear moments gives rise to a magnetic moment.

For a magnetic moment M in a uniform and static magnetic field H the equation of motion (Slichter, 1992) is

$$\frac{d\mathbf{M}}{dt} = \gamma[\mathbf{M} \times H] \tag{23}$$

The solutions to this equation are:

$$\begin{aligned} M_y(t) &= M_y(0)\cos(2\pi ft) + M_x(0)\sin(2\pi ft) \\ M_x(t) &= M_x(0)\cos(2\pi ft) - M_y(0)\sin(2\pi ft) \\ M_z(t) &= M_z(0) \end{aligned} \tag{24}$$

where x, y, and z refer to directions in the laboratory frame and the magnetic field is in the z direction. The frequency f is given by $\gamma|H|$. For typical magnetic fields of 1 to 30 T, the frequency is in the radiofrequency (rf) range, 1 MHz to 1 GHz.

The motion of the nuclear magnetization is precession about the magnetic field at the frequency γH. If the precessing magnetization occurs in a coil, a voltage is induced in the coil with the frequency f. The technically simple task of measuring the frequency of the induced voltage and a knowledge of the gyromagnetic ratio, γ, gives a measurement of $|H|$. The gyromagnetic ratio (γ) of a proton is 42.5759 [MHz/T]; it is known to an accuracy of better than 10^{-7}.

In the case of a slightly inhomogeneous field, the equations given above become

$$\begin{aligned} M_y(t) &= [M_y(0)\cos(2\pi ft) + M_x(0)\sin(2\pi ft)]G(t) \\ M_x(t) &= [M_x(0)\cos(2\pi ft) - M_y(0)\sin(2\pi ft)G(t)] \\ M_z(t) &= M_z(0) \end{aligned} \qquad (25)$$

where $G(t)$ is a function expressing the fanning out of the xy magnetization due to the distribution of magnetic fields in this inhomogeneous field. $G(t)$ is a direct measurement of the magnetic field homogeneity in the sample volume.

In the case of a time-varying magnetic field, the frequency given by $\gamma|H|$ will be time dependent. Slowly varying magnetic fields will be seen as a time dependence in the frequency f.

There are two commonly employed interferometric techniques to detect NMR signals: pulse and continuous wave (CW; Fukushima and Roeder, 1981). Both techniques make use of high-stability rf sources, with a stability of 10^{-8}/day.

In pulse NMR, the sample is irradiated with a short, intense rf pulse at the rf source frequency to nutate the magnetization away from the z axis and establish the precessing magnetization. Following the rf pulse, the voltage induced in the coil is detected, amplified, and mixed with the signal from the frequency source to yield a signal at the difference frequency, $f_{rf} - \gamma|H|$ (see NUCLEAR MAGNETIC RESONANCE IMAGING). This signal is digitized and analyzed, providing information about the magnitude, homogeneity, and time stability of the magnetic field.

In CW NMR, the sample is continuously irradiated with a low-level rf signal to maintain a precessing component of the magnetization and an audiofrequency magnetic field to modulate the magnetic field. Two coils aligned in quadrature, or balanced coils, are used to remove or reduce the irradiating rf signal. The resulting signal from the precessing magnetization is amplified and mixed with the rf source signal to yield a signal that is then further demodulated in a lock-in amplifier. The lock-in amplifier output signal can be analyzed for information about the magnitude, homogeneity, and time stability of the magnetic field.

The biggest limitation to using NMR in measuring magnetic fields is that the field must be reasonably homogeneous and stable. The spatial variations, field gradients, temporal variations, and field fluctuations must be less than 0.01%. NMR is not usually done in fields less than 1 T because the signal intensity is proportional to the square of the field. There is no limit to the largest fields that can be measured by NMR. The technical problems of the high frequencies due to large fields are circumvented by choosing nuclei with a lower γ.

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